Supplementary Material

Reproducible Low-Dimensional Structure Can Fail a Causal-State Test in Neural Training

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1 Purpose and claim boundary

This supplement documents the frozen experimental design, confirmation firewall, numerical results, protocol history, and reproducibility artifacts underlying the two-page student abstract. The main-paper inference is based on the final fresh-cohort experiment only: 20 discovery tasks and 30 untouched confirmation tasks in a controlled clean-learning ResNet18-GN/AdamW system. Earlier experiments are summarized only as design history; their observations are not pooled into the primary estimator.

The primary scientific conclusion is deliberately narrow: within the tested system, low-dimensional and/or independently reproducible response structure is not sufficient evidence for a causal state when the proposed coordinate fails to predict the future displacement caused by a hidden-state intervention. A separate positive claim establishes unusually high functional potency of historical AdamW first-moment memory relative to a matched history-orthogonal random panel. No claim is made here about architectural universality, natural-data generality, or a universal low-dimensional law of neural learning.

2 Operational causal-state criterion

At a training checkpoint let s_t denote the complete trainer state and let s'_t denote a branch in which the network parameters and buffers are unchanged but AdamW first-moment memory is surgically modified. The primary caused-response target is

$$\Delta Y_h = Y_{t+h}(s'_t) - Y_{t+h}(s_t),$$

with $h = 16$ for the held-out future endpoint. Short-horizon intervention responses at $h \in \{1, 2, 4\}$ are used to construct candidate coordinates. The primary causal-state requirement is stronger than ordinary absolute prediction: a discovery-frozen decoder of the proposed coordinate must predict ΔY_{16} on untouched tasks better than the zero-change predictor $\widehat{\Delta Y} = 0$. We report causal gain as

$$G = 1 - \frac{\mathrm{MSE}(\widehat{\Delta Y}, \Delta Y)}{\mathrm{MSE}(0, \Delta Y)},$$

so $G = 0$ matches zero-change, $G > 0$ improves on it, and $G < 0$ is worse.

The zero-change comparator is physically meaningful. If the hidden intervention has no predictable downstream effect, predicting no caused displacement is correct even when a richer observational model performs poorly. This avoids declaring causal success merely because a candidate coordinate beats a complicated but weak baseline.

3 Controlled learning system

The final experiment keeps the clean synthetic system fixed from the preceding protocol family:

- model: ResNet18 with GroupNorm, scale 2;
- optimizer: AdamW, learning rate 1.2×10^{-3}, weight decay 10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.999$;
- training: 500 optimizer updates, batch size 128;
- controlled task parameters: nuisance dimension $K = 32$, task-relevant dimension $r = 4$, clean labels ($\eta = 0$);
- recipient checkpoint: 70% of training; historical donor checkpoint: 40%;
- context horizons: $h = 1, 2, 4$; primary future horizon: $h = 16$.

The research cohort contains discovery task IDs 3000–3019 and confirmation IDs 3100–3129. Task identity hashes are checked for uniqueness. No earlier response table, fitted coordinate, decoder, or outcome is loaded into the final estimators.

4 Independent query families

Three physical query families are generated from independent noise streams as locally calibrated signed parameter perturbations applied at matched target modules. Q_A and Q_B provide independent short-horizon response vectors. On discovery tasks, regularized CCA identifies a shared coordinate from these matched views; a discovery-frozen linear decoder maps that coordinate to the physically disjoint Q_T response at $h = 16$. Confirmation applies the frozen transformations without refitting. The full query bank must satisfy a pre-specified geometric-independence support gate before outcome-bearing measurements are admitted. In the evaluable confirmation cohort, the maximum absolute direction cosine across Q_A, Q_T, Q_B is 0.14618, below the frozen 0.15 limit.

Every signed physical query also passes locality checks. Across confirmation branches, the minimum signed secant cosine is 0.9999982 and the maximum signed relative secant-norm drift is 0.0003585, compared with frozen requirements of at least 0.98 and at most 0.25, respectively.

5 Function-preserving optimizer-state interventions

At the recipient checkpoint, the model parameter and buffer state is held fixed and only AdamW first-moment memory (`exp_avg`) is modified. The primary history direction is formed from the recipient and earlier donor first moments. Random control directions are constructed independently under the frozen protocol. Nonzero branches must satisfy all of the following before contributing to a scientific claim:

- identical model parameters and buffers at the intervention cut;
- maximum instantaneous logit difference exactly zero;
- intended first-moment field changed;
- second moments, optimizer step counters, other state fields, and parameter groups unchanged;
- hidden-intervention locality: minimum signed secant cosine at least 0.98 and relative norm drift at most 0.25.

All 210 confirmation branches pass the foundational invariance/isolation checks. The observed minimum hidden secant cosine is 0.9999983 and maximum hidden relative secant-norm drift is 0.0003233.

6 Discovery/confirmation firewall and inference unit

The analysis is deliberately sequential. Discovery fixes representation construction, dimension comparison, regularization and decoder choices, thresholds, and the claim decomposition. Confirmation outcomes are evaluated only after this lock. The scientific inference unit is the independent task/replicate, never individual response rows. Reported confidence intervals are task-level bootstrap intervals; directional tests use task-level sign flips. This prevents pseudo-replication from the many response measurements generated inside each task.

The experiment does not use a master majority vote. Three logically distinct claims are reported independently:

1. between-task predictive phenotype;
2. within-task causal state;
3. optimizer anisotropy.

A positive optimizer-anisotropy result cannot rescue a failed scalar-state claim.

7 Complete final confirmation results

7.1 Between-task predictive phenotype

The discovery-selected effective dimension is one and remains non-inferior to its higher-dimensional challenger on confirmation. Absolute future prediction passes against a rich observational baseline under both independent query families:

Metric	Estimate	95% CI
Prediction improvement, Q_A	0.11410	[0.03145, 0.19497]
Prediction improvement, Q_B	0.10473	[0.01736, 0.18614]

Both corresponding one-sided task sign-flip tests pass ($p = 0.00555$ and 0.01299). Scalar sufficiency criteria also pass. However, the independently reconstructed query views fail to identify the same between-task ordering: correlation $r = 0.20284$ with 95% CI $[-0.18428, 0.55984]$. Therefore the full between-task claim is false despite low dimension, future prediction, and sufficiency.

7.2 Within-task causal-state test

Independent short-horizon reconstructions are strongly consistent within task:

$$\bar{r} = 0.77653, \qquad 95\%\ \text{CI} = [0.70016, 0.84492],$$

with positive agreement on all 30 confirmation tasks. The one-dimensional representation again remains non-inferior to its discovery-selected challenger.

Against the rich observational/treatment baseline, the scalar appears successful for centered future effects: improvements are 0.15741 (95% CI [0.07461, 0.23769]) for Q_A and 0.15683 (95% CI [0.07354, 0.23259]) for Q_B. The direct causal comparator reverses that conclusion. Relative to predicting zero caused change, improvements are

$$-0.02177\ [-0.03592, -0.00905] \quad (Q_A),$$

$$-0.02255\ [-0.03761, -0.00909] \quad (Q_B).$$

Both pre-specified positive-effect tests fail. A confirmation-only random-B intervention family also fails the zero-change criterion under both query views (means -0.01942 and -0.02089, with both 95% intervals below zero). Family-wise caused-contrast prediction likewise fails even though scalar-sufficiency tests pass. The frozen within-task causal-state claim is therefore false.

7.3 Optimizer anisotropy

The historical AdamW first-moment direction is compared with eight Gaussian control directions under a deliberately history-orthogonal null. Each control is normalized to the history-axis norm, projected into the orthogonal complement of history (and sequentially orthogonalized within the panel), and calibrated to the same prospective one-step logit-RMSE target. The reported potency ratio is therefore the random-direction first-moment displacement divided by the history-direction displacement needed to produce the same functional effect. The pre-specified history-versus-orthogonal-control potency claim passes on all 30 confirmation tasks:

Contrast	Median ratio	95% CI
History / median orthogonal control	90.9267×	[75.9508, 94.5708]
History / best of 8 orthogonal controls	70.7891×	[58.9567, 82.0031]

The frozen minimum requirements were 10× for the median-random ratio (CI lower bound at least 5×), 3× for the best-random ratio (CI lower bound at least 1.5×), and at least 80% of tasks above a 5× task-level floor. The observed fraction above the task-level floor is 1.0. Because the random panel is explicitly projected away from history, this result should be interpreted as extreme potency relative to a matched history-orthogonal null; it is not evidence that history dominates unrestricted random directions drawn from the entire optimizer-state space.

8 Why the three outcomes matter together

The final experiment separates state-like properties rather than treating "low dimensionality" as one monolithic phenomenon. The between-task representation is compact and predictively useful but not independently identifiable across query views. The within-task representation solves the query-identity problem but does not predict the caused future displacement better than zero. In contrast, the historical optimizer direction is exceptionally high-gain relative to random state directions. These outcomes motivate a shift from asking whether one scalar summarizes training to asking how functional control is geometrically organized in optimizer memory.

Importantly, the anisotropy result is not used as a post hoc rescue of either state claim. The final claims are separately frozen and separately reported.

9 Support-gate completion and provenance

The original confirmation campaign completed and sealed tasks 3100–3126, then stopped before final claim evaluation because the proposed physical query bank for task 3127 violated the already-frozen independence support gate (maximum absolute cosine $0.201394 > 0.15$). This was a support-geometry failure, not a scientific outcome.

A prospective completion rule preserved the 27 sealed tasks and the frozen discovery estimators. Before reading partial confirmation outcomes, it generated a deterministic whole-cohort query plan by rejection sampling from the same proposal distribution, accepting the first candidate bank whose maximum pairwise absolute cosine satisfied the original 0.15 support

threshold. Existing tasks 3100–3126 replayed candidate 0; task 3127 required candidate 1; 3128–3129 retained candidate 0. Only the three missing tasks were generated, after which the unchanged confirmatory analysis was evaluated once on all 30 tasks. The final completion protocol hash is

`89d065fb5d81d316667742c587edd1cdcf0b69c80a1e3a0a5adf9e4dafe434e1`.

The stopped parent run, support-only qualification rule, and completion evidence are preserved in the accompanying audit files.

10 Earlier frozen evidence as design history

The main paper does not pool these experiments into its estimator, but they explain why progressively stronger tests were designed.

Experiment	Frozen outcome and design implication
Nuisance-dimension pilot	Increasing nuisance dimension produced essentially no convincing interventional-state effect.
Task-relevant-dimension control	A reproducible one-dimensional intervention geometry was detected, yet held-out predictive improvement was strongly negative. This established that low-dimensional geometry alone is insufficient.
Objective-conflict pilot	Objective-conflict manipulation produced promising one-dimensional pilot evidence, but inferential confirmation was insufficient.
Pressure-emergence confirmation	The objective-conflict manipulation worked physically, but the pre-registered pressure-emergence hypothesis failed. Unexpectedly, clean learning showed the strongest one-dimensional predictive organization and approximate closure.
Causal-abstraction confirmation	A stronger causal-abstraction protocol failed overall. Function invariance, optimizer-field isolation, locality, one-dimensional adequacy, held-out prediction, and sufficiency passed; independent-query identifiability, controllability, unseen-intervention transport, and caused-contrast prediction failed.
Final decomposition	Decomposed the claims on fresh tasks: between-task predictive phenotype false; within-task causal state false; optimizer anisotropy true. This is the evidence used in the main abstract.

The architecture preflight in earlier work established common ResNet/ViT training and locality support only; it did not confirm cross-architecture transport. No architecture-generalization claim is made in this submission.

11 Reproducibility and provenance

The supplemental ZIP includes machine-readable frozen summaries, the final confirmation notebook, audit records, and a provenance manifest. Key identities include:

- Final confirmation-completion protocol hash: `89d065fb5d81d316667742c587edd1cdcf0b69c80a1e3a0a5adf9e4dafe434e1`;
- Interrupted parent result ZIP SHA-256: `ce32e9d26899ddcb9b32669e9fe67bb730cc1972f243a0490562edc54b39c947`;
- final completion result ZIP SHA-256 reported by the run: `a017f8ed47b68bbff089d2b4bb823f469508340cbff3fd35f1667d0642ba2729`;
- Earlier causal-abstraction protocol hash: `54d2dd5eaf1937b2457c2e7748d2bec524a5db2ff9ed2f728f8a0d9fa53a5bf5`.

The included `provenance_manifest.json` records SHA-256 hashes for the files packaged with this supplement.

12 Limitations and prospective work

The strongest limitation is external validity. The final claim concerns controlled synthetic clean-learning tasks, one ResNet18-GN architecture family, AdamW, one checkpoint geometry, and one specified family of local interventions. The experiment does not establish natural-data generality or transport across optimizers and architectures. It also does not prove that optimizer control is globally low rank.

Prospective work therefore asks a narrower mechanistic question: whether the local map from first-moment perturbations to future functional changes has a low-rank high-gain subspace and whether historical optimizer memory preferentially aligns with it. These experiments were designed after the Stage-6 outcome and are not used as evidence here.

References

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